

Observe the equations for $G_{3,11}$ and $G_{4,10}$ further give $G_{3,11} = -v$, where v solves the equation

$$v = \phi G_{3,3} G_{11,11} \bar{\text{tr}} \Sigma_1 (v \Sigma_1 + \lambda B) (\lambda I_d + G_{3,3} \Sigma_1)^{-1} (\lambda I_d + G_{11,11} \Sigma_1)^{-1}. \quad (73)$$

Now, let e be the unique non-negative solution to the equation

$$e = \frac{1}{1 + \phi \bar{\text{tr}} \Sigma_1 (\lambda I_d + e \Sigma_1)^{-1}}. \quad (74)$$

It is easy to see that we must have $G_{3,3} = G_{11,11} = e$ and

$$\begin{aligned} r^{(4)}/d &= \frac{G_{1,9}}{\lambda} = \bar{\text{tr}} A \Sigma_1 (eB - vI_d) (\lambda I_d + \Sigma_1)^{-2} \\ &= e^{-1} \bar{\text{tr}} AB \Sigma_1 (\kappa I_d + \Sigma_1)^{-2} - v e^{-2} \bar{\text{tr}} A \Sigma_1 (\kappa I_d + \Sigma_1)^{-2} \\ &= \frac{\kappa}{\lambda} \bar{\text{tr}} AB \Sigma_1 (\kappa I_d + \Sigma_1)^{-2} - \frac{v \kappa^2}{\lambda^2} \bar{\text{tr}} A \Sigma_1 (\kappa I_d + \Sigma_1)^{-2}, \end{aligned} \quad (75)$$

where $\kappa := \lambda/e$. Furthermore, v defined earlier now satisfies

$$\begin{aligned} v &= \phi e^2 \bar{\text{tr}} \Sigma_1 (v \Sigma_1 + \lambda B) (\lambda I_d + e \Sigma_1)^{-2} \\ &= \phi \bar{\text{tr}} \Sigma_1 (v \Sigma_1 + \lambda B) (\kappa I_d + \Sigma_1)^{-1}. \end{aligned}$$

Solving for v gives

$$v = \frac{\phi \lambda \bar{\text{tr}} B \Sigma_1 (\kappa I_d + \Sigma_1)^{-2}}{1 - \phi \bar{\text{tr}} \Sigma_1^2 (\kappa I_d + e \Sigma_1)^{-2}} \simeq \frac{\lambda \text{tr} B \Sigma_1 (\kappa I_d + \Sigma_1)^{-2}}{n - \text{df}_2(\kappa)}.$$

In particular, if $B = \Sigma_1$ and $A = I_d$, then

$$v = \frac{\lambda \text{df}_2(\kappa)}{n - \text{df}_2(\kappa)},$$

and so we must have

$$\begin{aligned} r^{(4)}/d &= \frac{G_{1,9}}{\lambda} = \frac{\kappa}{\lambda} \bar{\text{tr}} \Sigma_1^2 (\kappa I_d + \Sigma_1)^{-2} - \frac{v \kappa^2}{\lambda^2} \bar{\text{tr}} \Sigma_1 (\kappa I_d + \Sigma_1)^{-2} \\ &= \frac{\kappa}{\lambda} \frac{1}{d} \text{df}_2(\kappa) - \frac{\kappa^2}{\lambda} \frac{1}{d} \text{tr} \Sigma_1 (\kappa I_d + \Sigma_1)^{-2} \cdot \frac{\text{df}_2(\kappa)}{n - \text{df}_2(\kappa)} \\ &= \frac{\kappa}{\lambda} \frac{1}{d} \text{df}_2(\kappa) - \frac{\kappa}{\lambda} \frac{1}{d} \cdot (\text{df}_1(\kappa) - \text{df}_2(\kappa)) \cdot \frac{\text{df}_2(\kappa)}{n - \text{df}_2(\kappa)} \\ &= \frac{\kappa}{\lambda} \frac{1}{d} (n - \text{df}_1(\kappa)) \cdot \frac{\text{df}_2(\kappa)}{n - \text{df}_2(\kappa)} \\ &\simeq \frac{n}{d} \frac{\text{df}_2(\kappa)}{n - \text{df}_2(\kappa)} \simeq \frac{1}{\phi} \frac{\text{df}_2(\kappa)}{n - \text{df}_2(\kappa)}, \end{aligned} \quad (76)$$

where, in the last 2 steps we have made use of the following identities which follow from the definition of κ

$$\begin{aligned} \kappa - \lambda &\simeq \frac{\kappa \text{df}_1(\kappa)}{n}, \\ \kappa \text{tr} \Sigma_1 (\kappa I_d + \Sigma_1)^{-2} &= \text{df}_1(\kappa) - \text{df}_2(\kappa). \end{aligned}$$

We deduce that the variance term in the bias-variance decomposition of the test error is given by

$$\text{Var} = \sigma^2 \frac{1}{n} r^{(4)} \simeq \sigma^2 \frac{\text{df}_2(\kappa)}{n - \text{df}_2(\kappa)} = \sigma^2 u = \sigma^2 \frac{\text{df}_2(\kappa)/n}{1 - \text{df}_2(\kappa)/n}. \quad (77)$$

Let us now compute the limiting value of $r^{(4)}$ for any values of the proportions $p_1, p_2 \in (0, 1)$ with $p_1 + p_2 = 1$. The fixed-point equations defining $G_{1,9}$ now become

$$G_{1,9} = p_1 \bar{\text{tr}} A \Sigma_1 S (\lambda I_d + p_1 G_{2,2} \Sigma_1 + p_2 G_{6,6} \Sigma_2)^{-2}, \quad (78)$$

$$\text{with } S := \lambda (G_{2,2} B + G_{2,10} I_d) + p_2 (e_2 G_{2,10} - e_1 G_{6,13}) \Sigma_2, \quad (79)$$

$$G_{2,2} = e_1, \quad (80)$$

$$G_{6,6} = e_2, \quad (81)$$

$$G_{2,10} = G_{3,11} = \frac{\phi e_1^2 G_{4,10}}{\lambda}, \quad (82)$$

$$G_{6,13} = G_{7,14} = \frac{\phi e_2^2 G_{8,13}}{\lambda}, \quad (83)$$

$$G_{8,13} = \lambda \bar{\text{tr}} \Sigma_2 (\lambda B - p_1 G_{3,11} \Sigma_1 - p_2 G_{7,14} \Sigma_2) (\lambda I_d + p_1 G_{3,3} \Sigma_1 + p_2 G_{7,7} \Sigma_2)^{-2}, \quad (84)$$

$$G_{4,10} = \lambda \bar{\text{tr}} \Sigma_1 (\lambda B - p_1 G_{3,11} \Sigma_1 - p_2 G_{7,14} \Sigma_2) (\lambda I_d + p_1 G_{3,3} \Sigma_1 + p_2 G_{7,7} \Sigma_2)^{-2}, \quad (85)$$

where $e_1 \geq 0$ and $e_2 \geq 0$ solve the following system of equations

$$e_1 = \frac{1}{1 + \phi \bar{\text{tr}} \Sigma_1 (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-2}}, \quad (86)$$

$$e_2 = \frac{1}{1 + \phi \bar{\text{tr}} \Sigma_2 (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-2}}. \quad (87)$$

Furthermore, we deduce that $G_{6,13} = -v_2$ and $G_{2,10} = -v_1$, where v_1 and v_2 solve the equations

$$v_1 = \phi e_1^2 \bar{\text{tr}} \Sigma_1 (p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2 + \lambda B) (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-2}, \quad (88)$$

$$v_2 = \phi e_2^2 \bar{\text{tr}} \Sigma_2 (p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2 + \lambda B) (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-2}. \quad (89)$$

Putting things together gives the formula for $r^{(4)}$ proposed in Proposition 2. $\square$

In particular, taking $p_2 \rightarrow 0$ (i.e $p_1 \rightarrow 1$) recovers the formula as a special case.

Computing $r^{(3)}$. A minimal linear pencil for the corresponding random matrix $R = AM_1(M + \lambda I_d)^{-1}B(M + \lambda I_d)^{-1}M_1$ is a 17×17 block matrix Q such that $R = Q^{-1}[1, 16]$. This gives

$$r^{(3)}/d \simeq G_{1,16},$$

where $G = (\text{id} \otimes \mathbb{E} \bar{\text{tr}})[Q^{-1}] \in M_{17}(\mathbb{C})$. The fixed-point equations that determine $G_{1,16}$ are

$$\begin{aligned} G_{1,16} &= p_1 \bar{\text{tr}} A \Sigma_1 S (\lambda I_d + p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2)^{-2} \\ \text{with } S &:= p_1 e_1^2 (\lambda B - p_2 G_{6,13} \Sigma_2) \Sigma_1 - G_{2,10} (\lambda I_d + p_2 e_2 \Sigma_2)^2, \\ G_{7,14} &= G_{6,13} = -v_2, \\ G_{3,11} &= G_{2,10} = -v_1. \end{aligned}$$

We deduce the formula given in Proposition 2. In particular, taking the limit $p_2 \rightarrow 0$ (i.e $p_1 \rightarrow 1$) gives

$$\begin{aligned} \bullet \tilde{S} &\simeq e_1^2 B \Sigma_1 + \lambda v_1 I_d = e_1^2 B \Sigma_1 + \lambda^2 u_1 I_d, \\ \bullet v_1 &= \phi e_1^2 \bar{\text{tr}} \Sigma_1 (v_1 \Sigma_1 + \lambda B) (e_1 \Sigma_1 + \lambda I_d)^{-2} = \phi \bar{\text{tr}} \Sigma (v_1 \Sigma_1 + \lambda B) (\Sigma + \kappa_1 I_d)^{-2}, \text{ i.e} \\ u_1 &= \frac{v_1}{\lambda} = \frac{\phi \bar{\text{tr}} B \Sigma_1 (\Sigma_1 + \kappa_1 I_d)^{-2}}{1 - \phi \bar{\text{tr}} \Sigma_1^2 (\Sigma_1 + \kappa_1 I_d)^{-2}} \simeq \frac{\text{tr } B \Sigma_1 (\Sigma_1 + \kappa_1 I_d)^{-2}}{n - \text{df}_2^{(1)}(\kappa_1)}. \end{aligned} \quad (90)$$

Finally, recalling that $\kappa_1 = \lambda/e_1$ by construction, we get

$$\begin{aligned} r^{(3)} &\simeq d \cdot G_{1,16} = e_1^2 \text{tr } AB \Sigma_1^2 (e_1 \Sigma_1 + \lambda I_d)^{-2} + \lambda^2 u_1 \bar{\text{tr}} A \Sigma_1 (e_1 \Sigma_1 + \lambda I_d)^{-2} \\ &= \text{tr } AB \Sigma_1^2 (\Sigma_1 + \kappa_1 I_d)^{-2} + \frac{\lambda^2 u_1}{e_1^2} \text{tr } A \Sigma_1 (\Sigma_1 + \kappa_1 I_d)^{-2} \\ &\simeq \text{tr } AB \Sigma_1^2 (\Sigma_1 + \kappa_1 I_d)^{-2} + \kappa_1^2 \text{tr } A \Sigma_1 (\Sigma_1 + \kappa_1 I_d)^{-2} \cdot \frac{\text{tr } B \Sigma_1 (\Sigma_1 + \kappa_1 I_d)^{-2}}{n - \text{df}_2^{(1)}(\kappa_1)}. \end{aligned}$$

Computing $r^{(2)}$. A pencil for the relevant matrix $R = \lambda^2 A(M + \lambda I_d)^{-1} B(M + \lambda I_d)^{-1}$ has minimal linear pencil Q of size 15×15 , where $R = Q^{-1}[1, 8]$. We deduce that $r^{(2)}/d = \mathbb{E} \bar{\text{tr}} R / \lambda^2 = G_{1,8}/\lambda^2$, where $G = (\text{id} \otimes \mathbb{E} \bar{\text{tr}}) Q^{-1} \in M_{15}(\mathbb{C})$. The fixed-point equations defining $G_{1,8}$ are given by

$$G_{1,8} = \lambda \bar{\text{tr}} AS(p_1 G_{2,2} \Sigma_1 + p_2 G_{5,5} \Sigma_2 + \lambda I_d)^{-2}, \quad (91)$$

$$\text{with } S = \lambda B - p_1 G_{2,9} \Sigma_1 - p_2 G_{5,12} \Sigma_2, \quad (92)$$

$$G_{2,2} = e_1, \quad (93)$$

$$G_{5,5} = e_2, \quad (94)$$

$$G_{2,9} = G_{3,10} = \frac{\phi e_1^2 G_{4,9}}{\lambda}, \quad (95)$$

$$G_{5,12} = G_{6,13} = \frac{\phi e_2^2 G_{7,12}}{\lambda}, \quad (96)$$

$$G_{4,9} = -\lambda \bar{\text{tr}} \Sigma_1 (\lambda B - p_1 G_{3,10} \Sigma_1 - p_2 G_{6,13} \Sigma_2) (p_1 G_{2,2} \Sigma_1 + p_2 G_{5,5} \Sigma_2 + \lambda I_d)^{-2}, \quad (97)$$

$$G_{7,12} = -\lambda \bar{\text{tr}} \Sigma_2 (\lambda B - p_1 G_{3,10} \Sigma_1 - p_2 G_{6,13} \Sigma_2) (p_1 G_{2,2} \Sigma_1 + p_2 G_{5,5} \Sigma_2 + \lambda I_d)^{-2}. \quad (98)$$

Based on previous calculations, we deduce that $G_{2,9} = -v_1$ and $G_{5,12} = -v_2$, and so

$$r^{(2)} \simeq d \cdot \frac{G_{1,8}}{\lambda^2} = \frac{1}{\lambda} \text{tr} A(p_1 v_1 \Sigma_1 + p_2 v_2 \Sigma_2 + \lambda B) (p_1 e_1 \Sigma_1 + p_2 e_2 \Sigma_2 + \lambda I_d)^{-2} = \text{tr} A \tilde{L} K^{-2},$$

as claimed. This completes the proof of Proposition 2. $\square$

D.4 Proof of Proposition 3

In Section F.1 we will establish a more general result which implies Proposition 3 as a special case.

E Proof of Theorem 1 and Corollaries

Let us note that the results in Bach (2023) were obtained in a two-stage approach, where random matrix theory is applied on the raw (unquenched test error $\|\hat{w} - w_1\|_\Sigma^2$ with the projection matrix treated like a deterministic matrix, and then RMT is done one more on the resulting expressions but now treating S as random. The case general case $p_2 \in (0, 1)$ is much more difficult; the key technical difficulty can be pinned down to the problem of obtaining analytic deterministic equivalents for the trace of the and derivatives of the resolvent of a sum of random matrices. To circumvent this, we employ the tools of operator-valued free probability theory.

E.1 Proof of Theorem 1

From Proposition 1 and 2 applied with $\Sigma_1 = \Sigma_2 = \Sigma$, we know that

$$E_{\text{test}}(\hat{f}_{CL}) = V_1 + B_1, \text{ with}$$

$$\begin{aligned} V_1 &\simeq \sum_{j=1}^2 p_j \sigma_j^2 \frac{1}{n} \text{tr} \Sigma_k D_{j,k} K^{-2} = \sum_{j=1}^2 p_j \sigma_j^2 \frac{\kappa}{\lambda} \cdot \frac{1}{n} \text{tr} \Sigma (\Sigma - \kappa u I_d) (\Sigma + \kappa I_d)^{-2} \\ &= \sigma^2 \frac{\kappa}{\lambda} \cdot \frac{1}{n} \text{tr} \Sigma (\Sigma - \kappa u I_d) (\Sigma + \kappa I_d)^{-2}, \\ B_1 &= p_2 \text{tr} \Delta \Sigma_2 C_{2,1} K^{-2} + \lambda^2 \text{tr} \Gamma L'_1 K^{-2} \\ &= p_2 \text{tr} \Delta \Sigma (p_2 (1 + p_1 u) \Sigma^2 + u (p_1 \Sigma + \kappa I_d)^2) (\Sigma + \kappa I_d)^{-2} + \kappa^2 (u + 1) \text{tr} \Gamma \Sigma (\Sigma + \kappa I_d)^{-2}. \end{aligned}$$